

ORGANIC CHEMISTRY — Complete Study Guide

Dr. Saif Soliman · G12 · S'26 · Sessions 13–20

1. FOUNDATIONS (S13)

Carbon & Bonding

- Group 4A, 4 valence e[−], 4 covalent bonds
- sp³: tetrahedral 109.5°. sp²: 120°. sp: 180°
- Single (σ) · double (σ+π) · triple (σ+2π)

Inorganic C-Exceptions (not organic)

CO, CO₂, CO₃^{2−}, HCO₃[−], CN[−], CNO[−], CNS[−]

Wöhler 1828: NH₄CNO $\xrightarrow{\Delta}$ urea $\Rightarrow$ Vital Force Theory destroyed.

Functional Group Priority (suffix = highest)

COOH	-oic acid	CHO	-al
COOR	-oate	C=O	-one
CONH ₂	-amide	OH	-ol
C=C	-ene	C≡C	-yne

Priority: COOH > COOR > CONH₂ > CHO > C=O > OH > NH₂ > C=C > C≡C > X

Hydrocarbon Classes

Class	Formula	Bond	Activity
Alkane	C _n H _{2n+2}	σ	Inactive
Alkene	C _n H _{2n}	σ+π	Active
Alkyne	C _n H _{2n−2}	σ+2π	Most active
Cycloalkane	C _n H _{2n}	σ	Low
Cycloalkene	C _n H _{2n−2}	σ+π	Active

IUPAC Naming — 5 Steps

- Longest chain containing the key F.G. = parent
- Number from the end nearest first substituent (or C=C/C≡C)
- Assign locants to all substituents
- Write as one word; hyphens before locants; commas between locants; alphabetise (iso- counts; sec-/tert- do not)
- Complex groups in parentheses

C/H Classification

1°C bonded to 1C · 2°to 2C · 3°to 3C · 4°to 4C.
4°H does not exist.

2. ALKANES (S15)

Physical Properties

- State: C₁–C₄ gas · C₅–C₁₇ liquid · C₁₈⁺ solid
- BP:** straight → branched (same C#) — more surface area $\Rightarrow$ stronger VdW
- MP:** even-C > odd-C neighbours (symmetric packing); more branches $\Rightarrow$ lower MP
- Non-polar; immiscible H₂O; soluble CCl₄/ether

Chemical Reactions

I. Halogenation (Free Radical, UV/400°C)

Reactivity: F₂ > Cl₂ > Br₂ > I₂ Ease: 3°H > 2°H > 1°H

Mechanism:

Initiation: Cl₂ $\xrightarrow{h\nu}$ 2Cl• (homolytic)

Propagation: CH₄ + Cl• $\rightarrow$ HCl + CH₃• $\rightarrow$ +Cl₂ $\rightarrow$ CH₃Cl + Cl•

Termination: two radicals combine

II. Heating Reactions (no air)

- Cracking:** long chain $\rightarrow$ short alkane + alkene
- Pyrolysis:** CH₄ $\xrightarrow{1000C}$ C(black) + 2H₂
- Quenching:** 2CH₄ $\xrightarrow{1500C, rapid cool}$ HC≡CH + 3H₂
- Water gas:** CH₄ + H₂O $\xrightarrow{725C}$ CO + 3H₂
- Reforming (****):** C₆H₁₄ $\xrightarrow{Pt/\Delta}$ C₆H₆ + 4H₂

III. Other Reactions

Dehydrogenation: C₃H₈ $\xrightarrow{Cr_2O_3/500C}$ CH₃CH=CH₂ + H₂

Grignard: R–X + Mg $\xrightarrow{ether}$ RMgX; +CO₂+H₂O $\rightarrow$ RCOOH

Wurtz: 2R–X + 2Na $\xrightarrow{\Delta}$ R–R + 2NaX

Combustion (complete): C_nH_{2n+2} + (n + $\frac{n+1}{2}$)O₂ $\rightarrow$ nCO₂+(n+1)H₂O

Lab Prep of Alkanes

Dry distillation (soda lime = CaO/NaOH):

R–COONa + NaOH $\xrightarrow{CaO, \Delta}$ R–H + Na₂CO₃

Product alkane has **1 C less** than the acid salt.

3. CYCLIC HC & AROMATICITY (S14,S19)

Ring Stability vs Strain

Ring	Angle	H ₂ /Ni Temp	Stability
Cyclopropane	60°	180°C	Least
Cyclobutane	90°	200°C	Less
Cyclopentane	108°	320°C	Less
Cyclohexane	120°	No rxn	Most

Hückel's Rule — Aromatic if ALL 4:

(1) Cyclic (2) Planar, all sp² (3) Fully conjugated (4) 4n+2 πe[−] (n = 0, 1, 2, ...)

Aromatic: 2,6,10,14,... πe[−] Anti: 4,8,12,...

Benzene Bonds

All bonds intermediate (delocalized π): length & strength between single and double. Planar, sp².

Benzene Reactions

1. Addition (hard—delocalized π resists)

+3H₂/Pt $\rightarrow$ cyclohexane +3Cl₂/UV $\rightarrow$ C₆H₆Cl₆ (Gamaxane, insecticide)

2. Electrophilic Substitution

Type	Conditions	Adds
Nitration	HNO ₃ /H ₂ SO ₄ /50°C	–NO ₂
Sulfonation	H ₂ SO ₄ /50°C	–SO ₃ H
FC Alkylation	RCI/AlCl ₃ /50°C	–R
FC Acylation	RCOCl/AlCl ₃	–COR
Halogenation	Cl ₂ /FeCl ₃	–Cl/Br

Directing Groups

ortho/para: OH, NH₂, R (alkyl), X (halogens)

meta: NO₂, COOH, CHO, COR, SO₃H

Key Benzene Conversions

Benzene $\xrightarrow{FC, RCl}$ toluene $\xrightarrow{KMnO_4}$ benzoic acid
Benzene $\xrightarrow{Cl_2/FeCl_3}$ chlorobenzene $\xrightarrow{NaOH/300C}$ phe-nol
phenol $\xrightarrow{Zn/\Delta}$ benzene
Toluene + 3HNO₃/H₂SO₄ $\rightarrow$ TNT (2,4,6-trinitrotoluene)
Phenol + 3HNO₃/H₂SO₄ $\rightarrow$ picric acid (2,4,6-TNP)
3HC≡CH $\xrightarrow{Ni/\Delta}$ benzene (trimerization)

4. ALKENES (S15–S16)

Preparation

- Cracking of alkanes (cat/heat, no air)
- Dehydration of alcohols: Al₂O₃/Δ or H₂SO₄/180°C
- Dehydrogenation: Cr₂O₃/500°C

Addition Reactions

Symmetric Reagents

Hydrogenation: H₂/Ni/150–300°C $\rightarrow$ alkane; moles H₂ = no. π bonds

Halogenation: X₂/CCl₄ $\rightarrow$ dihalide (1 mol π bond = 1 mol Br₂); decolorizes Br₂

Asymmetric Reagents

Markovnikov: H⁺ to C with *more* H's; X[−] to C with *fewer* H's.

Basis: alkyl groups have +I effect $\Rightarrow$ stabilises carbocation.

HX (no peroxide) $\rightarrow$ Markovnikov product
HBr + H₂O₂ $\rightarrow$ **Anti-Markovnikov** (Br to less-substituted C)

Only HBr undergoes anti-Markov. (not HCl or HI)

Catalytic Hydration: H₂O/H₂SO₄/110°C $\rightarrow$ alcohol (Markovnikov)

Only **ethene** gives 1°alcohol by hydration; all others give 2°or 3°.

Elimination (reverse of addition)

Alkyl halide + alc. KOH $\rightarrow$ alkene + KX

Zaitsev: major product = more substituted alkene

Oxidation

No cleavage (cold dil. KMnO₄/OH[−]): $\rightarrow$ diol (Bayer's); violet discharged

Cleavage (hot KMnO₄/H⁺):

C of C=C	H count	Product
=CR ₂	0H	Ketone
=CHR	1H	RCOOH
=CH ₂	2H	CO ₂ +H ₂ O

Ozonolysis (reductive), O₃ then Zn/H⁺: CHR gives **aldehyde** (not acid)

Quick Conditions Reference

Reaction	Conditions
Hydrogenation	H ₂ /Ni/150–300°C
Halogenation	X ₂ /CCl ₄
HX Markov.	HX alone
HBr Anti-M.	HBr + H ₂ O ₂
Hydration	H ₂ O/H ₂ SO ₄ /110°C
Elimination	Alc. KOH
Substitution	Aq. KOH
Non-cleave ox.	Cold dil. KMnO ₄ /OH [−]
Cleavage ox.	Hot KMnO ₄ /H ⁺
Dehydr. $\rightarrow$ alkene	H ₂ SO ₄ /180°C
Dehydr. $\rightarrow$ ether	H ₂ SO ₄ /140°C

Polymerisation (Addition)

Unsaturated monomers + free radical (H₂O₂/O₂); π bond opens; same EF.

Ethene $\rightarrow$ PE · Propene $\rightarrow$ PP · CH₂=CHCl $\rightarrow$ PVC · CF₂=CF₂ $\rightarrow$ Teflon

5. ALKYNES (S17)

Preparation of Ethyne

CaCO₃ $\xrightarrow{\Delta}$ CaO+CO₂ · 3C+CaO $\xrightarrow{2000C}$ CaC₂+CO · CaC₂+2H₂O $\rightarrow$ **C₂H₂**+Ca(OH)₂

Reactions

- Hydrogenation:** +2H₂/Ni $\rightarrow$ alkane (alkyne $\rightarrow$ alkene $\rightarrow$ alkane)
- Halogenation:** +2Br₂ $\rightarrow$ 1,1,2,2-tetrabromoalkane
- HX (Markov.):** step 1 $\rightarrow$ vinyl halide; step 2 $\rightarrow$ **1,1-dihalide** (NOT 1,2)

Hydration—Tautomerism

Alkyne + H₂O $\xrightarrow{HgSO_4/H_2SO_4}$ enol (unstable) $\rightleftharpoons$ **keto** (stable)

Ethyne+H₂O $\rightarrow$ ethanal Propyne+H₂O $\rightarrow$ acetone (propanone)

Acidity of Terminal Alkynes

sp orbital (50%)s holds e[−] closer $\rightarrow$ most acidic C–H

pKa: alkyne 25 < alkene 44 < alkane 50 (lower = more acidic)

- CuSO₄/NH₄OH $\rightarrow$ **reddish-brown** ppt (cop-per acetylide)
- AgNO₃/NH₃ $\rightarrow$ **white** ppt (silver acetylide)
- Symmetric/internal alkynes (no terminal H): no reaction*

Trimerization: 3HC≡CH $\xrightarrow{\text{red-hot Ni}}$ **benzene**

Detection Test Summary

Test	Alkane	Alkene	Alkyne
Br ₂ /CCl ₄	red stays	disc.	disc.
KMnO ₄ /OH [−]	violet stays	disc.	violet stays
CuSO ₄ /NH ₄ OH	–ve	–ve	red-brn
AgNO ₃ /NH ₃	–ve	–ve	white ppt

6. ISOMERISM (S18)

Structural Isomers

- Chain:** different C skeleton; similar chemical properties
- Positional:** same skeleton + FG, different position
- Functional Group:** same MF, different FG (see table in S14)

Count: C₄H₁₀: 2 · C₅H₁₂: 3 · C₆H₁₄: 5

Stereoisomers

A. Geometrical (cis/trans)

Conditions: (1) restricted rotation (C=C or cycloalkane); (2) each C of double bond carries *two different* groups.

cis/Z: priority groups *same* side · trans/E: *opposite* side

Trans more stable (less steric repulsion)

CIP priority: I > Br > Cl > F > C > H; isopropyl > propyl

B. Optical Isomerism

Chiral centre = sp³ C bonded to **4 different** groups.
No. of optical isomers = 2ⁿ (n = chiral centres).

Meso form: has internal plane of symmetry ⇒ optically *inactive* despite chiral centres.
2,3-dibromobutane: 2 chiral C → 3 stereoisomers (2 enantiomers + 1 meso).
Racemic (dl): 50:50 enantiomers → optically inactive.

7. ALCOHOLS & PHENOL (S14, S19)

Classification

By OH count: mono- · di- (glycol) · tri- (glycerol) · poly- (sorbitol)
By carbinol C: 1°(2H) · 2°(1H) · 3°(0H)

Physical Properties

- Polar OH → H-bonds → miscible with H₂O, high BP
- BP: sorbitol > glycerol > EG > ethanol > methanol
- Monohydric: larger MM → lower miscibility (hydrophobic tail)
- More branches → lower BP

Alcohol vs Phenol — Key Differences

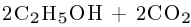
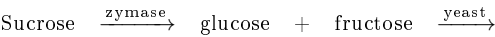
Why phenol is more acidic:
Alcohol (+I of R) → e[−] density on O ↑ → O–H strong → no reaction with NaOH.
Phenol (−I of ring) → e[−] on O ↓ → O–H weak → reacts with NaOH.

Reagent	Alcohol	Phenol
Na metal	alkoxide + H ₂	phenoxide + H ₂
NaOH	no rxn	Na-phenoxide
FeCl ₃	−ve	violet
Br ₂ /CCl ₄	−ve	white ppt

Oxidation of Alcohols

1° → (mild KMnO₄) aldehyde → (full) carboxylic acid
2° → (KMnO₄) ketone
3° → **no reaction** (no H on carbinol C); KMnO₄ stays violet
Iodoform (I₂/NaOH): +ve for CH₃–CH(OH)– only (2° with adjacent CH₃)

Fermentation



8. CARBOXYLIC ACIDS & ESTERS (S14, S20)

Carboxylic Acids

Acidity Rules

1. Inorganic > organic; HCOOH > CH₃COOH (MM↑ → acidity↓)
2. Aromatic > aliphatic (benzoic > acetic)
3. EWG (NO₂) on ring → acidity ↑; EDG (OH) → acidity ↓
4. Halo: F>Cl>Br; halogen closer to COOH → stronger acid

Order: HCl > benzoic > HCOOH > CH₃COOH > EtOH > phenol

Detection

Na₂CO₃ or NaHCO₃ → **effervescence** + **CO₂** (limewater milky)

Key Reactions

Reduction: RCOOH + 2H₂ $\xrightarrow{\text{CuCrO}_4/400C}$ R–CH₂OH (1° alcohol)

Esterification: RCOOH + HOR' $\xrightarrow{\text{H}_2\text{SO}_4/\Delta}$ RCOOR' + H₂O (reversible)

Decarboxylation: RCOONa + NaOH $\xrightarrow{\text{CaO}/\Delta}$ alkane + Na₂CO₃

BP order (same MM): Acid > Alcohol > Ester > Ether
Acids form **2 H-bonds** (dimers); alcohols 1; esters/ethers: only VdW.

Esters

Key Drug Esters

Aspirin (ASA): salicylic acid + ethanoic acid $\xrightarrow{\text{dry HCl}}$ acetylsalicylic acid

Oil of wintergreen: salicylic acid + methanol $\xrightarrow{\text{H}_2\text{SO}_4}$ methyl salicylate

Hydrolysis

Acidic (H₂SO₄/heat) → acid + alcohol
Alkaline/saponification (NaOH) → carboxylate salt + alcohol
Ammonolysis (NH₃) → amide + alcohol
Triglyceride + NaOH → glycerol + soap

Oils, Fats & Triglycerides

Oil = unsaturated fatty acids · Fat = saturated fatty acids
Glycerol + 3 fatty acids → triglyceride (ester bond)
Glycerol + HNO₃/H₂SO₄ → TNG (nitroglycerin, explosive/vasodilator)

9. POLYMERISATION (S16, S20)

Addition (homopolymer)

Unsaturated + free radical (H₂O₂) → polymer; same empirical formula; π opens.

Condensation (copolymer)

Two bifunctional monomers → polymer + H₂O (or HCl)

- **PET (Polyester):** terephthalic acid + 1,2-ethanediol → [–C₆H₄–COO–CH₂CH₂–O–]_n
- **Nylon 6,6:** 1,6-hexanedioic acid + 1,6-hexanediamine → polyamide + H₂O

10. MASTER CONVERSIONS

CH₄ → Benzene:
CH₄ $\xrightarrow{1500C}$ HC≡CH $\xrightarrow{3 \times Ni}$ Benzene
Butanoic acid → Propene:
Neutralise → NaButanoate $\xrightarrow{\text{CaO}}$ Propane $\xrightarrow{\text{Cr}_2\text{O}_3}$ Propene
Heptanoic acid → Benzene:
→ NaHeptanoate $\xrightarrow{\text{CaO}}$ Hexane $\xrightarrow{\text{Pt}/\Delta}$ Benzene + 4H₂
Ethanoic acid → Benzene:
 $\xrightarrow{\text{KMnO}_4}$ AcOH $\xrightarrow{\text{NaOH}}$ NaOAc $\xrightarrow{\text{CaO}}$ CH₄ $\xrightarrow{1500C}$ HC≡CH $\xrightarrow{Ni}$ Benz.
Ethanoic acid → Ethane:
 $\xrightarrow{\text{NaOH}}$ NaOAc $\xrightarrow{\text{CaO}}$ CH₄ $\xrightarrow{\text{Cl}_2/UV}$ CH₃Cl $\xrightarrow{2Na}$ C₂H₆
1° → 2° alcohol:
Dehydrate (180°C) → alkene $\xrightarrow{\text{H}_2\text{O}/\text{H}_2\text{SO}_4}$ 2° alcohol
2° → 1° alcohol:
Dehydrate → alkene $\xrightarrow{\text{HBr}/\text{H}_2\text{O}_2}$ R–Br $\xrightarrow{\text{aq. KOH}}$ 1° alcohol
Methyl formate from methanol only:
Split: (X) keep as CH₃OH; (Y) oxidise KMnO₄ → HCOOH
Mix X+Y + H₂SO₄ → HCOOCH₃

11. NAMED REACTIONS QUICK REF

Name	Description
Wöhler	NH ₄ CNO → urea; killed vital force theory
Bayer's Markovnikov	Alkene + cold KMnO ₄ /OH [−] → diol H to more-H C; X to fewer-H C
Zaitsev	Elim. major = more substituted alkene
Tautomerism	Enol ⇌ keto (keto more stable)
Ozonolysis	O ₃ /Zn/H ⁺ (red.) or O ₃ /H ₂ O ₂ (ox.)
Wurtz	2RX + 2Na → R–R + 2NaX
Grignard	RX + Mg → RMgX; +CO ₂ /H ₂ O → RCOOH
Saponification	Ester + NaOH → soap + alcohol
Friedel-Crafts	Benz + RCl or RCOCl/AlCl ₃ → sub.

FG Isomer Pairs (same MF)

Cycloalkane ↔ Alkene · Cycloalkene ↔ Alkyne · Alcohol ↔ Ether · Acid ↔ Ester · Aldehyde ↔ Ketone